A linear information bound for nonsymmetric convex classes under scalar-error measurability

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Abstract

We extend the $27n$ comparison of Krieg, Novak and Ullrich to the scalar-error measurability convention in AIM 568. A finite-dimensional measurable-majorant lemma supplies the bridge without assuming that the original adaptive rules or reconstruction are measurable. The source theorem supplies the geometric lower bound; the contribution here is this bridge.

Catalogue target. AIM 568 at repository revision `c4036871b0bf`. The package preserves its full statement in `statement.md`; the review and target comparison are linked from the package README.

1 Problem 568: closing the measurability gap

Throughout, $e_n^{\mathrm{ran}}(S,F)$ uses expected norm error and a strict budget of at most n queries in every realization, as required by AIM 568. The class $F \subset X$ is bounded and convex, without a symmetry assumption. Its measurability convention uses the norm p_F on $\operatorname{span}(F-F)$.

Krieg, Novak and Ullrich's October 2026 preprint [1] proves the $27n$ comparison with measurable adaptive decision rules and reconstruction. The catalogue retains only measurability of the scalar error, so the theorem cannot simply be quoted in its stated model. The result below supplies the missing bridge.

Theorem 1.1. *Under exactly the assumptions of Problem 568,*

$$e_{2n}^{\mathrm{det,non}}(S,F) \le e_{2n-1}^{\mathrm{det,non}}(S,F) \le a_{2n-1}(S,F) \le 27n\, e_n^{\mathrm{ran}}(S,F), \qquad n \ge 1, \tag{568.1}$$

where a_m is the minimal worst-case error of continuous affine algorithms whose linear part has rank at most m.

The numerical comparison constant is inherited from [1]. The contribution here is its extension to the larger admissibility class specified in the question.

*The solutions were found by AI. The submitted arguments and the repository audit are AI work; independent human review and formal verification are not claimed.

1.1 The catalogue measurability convention

Let X, Y be real Banach spaces, $S : X \to Y$ bounded and linear, and $F \subset X$ nonempty, bounded and convex. On $\operatorname{span}(F - F)$ define

$$p_F(v) = \inf\{t > 0 : v \in t(F - F)\}.$$

The input Borel structure is that of the metric $d_F(f, g) = p_F(f - g)$ (the one-point topology in the trivial case). Each randomized realization uses at most n adaptive continuous linear measurements. Only the scalar error $(f, \omega) \mapsto \|Sf - A_\omega(f)\|_Y$ must be jointly measurable. Its criterion is $\sup_{f \in F} \mathbb{E}\|Sf - A_\omega(f)\|_Y$, denoted $e_n^{\mathrm{ran}}(S, F)$ after minimization. The deterministic quantity $e_m^{\mathrm{det,non}}(S, F)$ minimizes the worst-case norm error over m fixed continuous linear measurements and arbitrary reconstruction. We prove the requested comparison with $C = 27$, in fact at budget $2n - 1$.

1.2 A measurable-majorant principle

Lemma 1.2. *Let $K \subset \mathbb{R}^d$ be compact, $T : K \to \mathbb{R}^q$ continuous, ν a Borel probability measure on K, and $g : K \to [0, \infty]$ Borel. Suppose an algorithm A, with entirely arbitrary adaptive decisions and decoder, uses at most k affine scalar measurements and satisfies*

$$\|T(x) - A(x)\|_2 \le g(x) \tag{B1}$$

outside a Borel ν-null set. Then, for every $\varepsilon > 0$, there is an algorithm B with at most k affine scalar measurements, Borel history-to-query maps and a Borel decoder, such that

$$\int_K \|T(x) - B(x)\|_2 \, d\nu(x) \le \int_K g \, d\nu + \varepsilon. \tag{B2}$$

The algorithm's own error in (B1) need not be measurable.

Proof. Choose a closed Euclidean ball C of radius R containing $T(K)$. Projecting a decoder onto C cannot increase its distance to any $T(x)$. We may therefore restrict all candidate decoders to C; their losses are bounded by $2R$.

For each $j \ge 0$, let $\mathcal{C}_j$ be the countable family of j-query policies whose affine query coefficients are rational-coefficient polynomials of the previous answers and whose decoder is the projection onto C of a rational-coefficient polynomial map. Each complete input-output map in this family is continuous on K.

We first establish, for every Borel probability λ on K,

$$\inf_{B \in \mathcal{C}_j} \int_K \|T - B\|_2 \, d\lambda = \inf_{\substack{B:\ j \text{ queries} \\ \text{Borel decisions and decoder}}} \int_K \|T - B\|_2 \, d\lambda. \tag{B3}$$

Start with a Borel policy and consider each of its *original* finite-history probability laws. By Lusin's theorem, its query coefficient map is continuous on a compact set of that history space having probability at least $1 - \delta$. Extend the coefficients continuously by coordinatewise Tietze extension. Do the same for the decoder, then project the extension onto C. On the event that every original history belongs to its designated compact set, the modified policy follows precisely the same history and gives the same output, by induction through the stages. The complement has probability at most $(j + 1)\delta$. Hence the change in mean loss is at most $2R(j + 1)\delta$.

The modified policy is continuous in its histories. Its reachable histories on compact K are compact. Approximate all coefficient maps and the decoder on compact boxes containing neighborhoods of those histories by rational-coefficient polynomials. Successive uniform

continuity through the finite observation recursion shows that sufficiently close approximations give uniformly close final outputs on K. Projection onto C is nonexpansive. This proves (B3); the reverse inequality is immediate.

Define

$$V_j(\lambda) = \inf_{B \in \mathcal{C}_j} \int_K \|T(x) - B(x)\|_2 \, d\lambda(x). \tag{B4}$$

For each fixed $B \in \mathcal{C}_j$, its loss is continuous and bounded on K, so its integral is continuous for the weak topology on $\mathcal{P}(K)$. Thus V_j is an infimum of countably many continuous functions, and is upper semicontinuous and Borel. By (B3), it is exactly the optimal Borel-policy average error.

We now prove $\int g \, d\nu \geq V_k(\nu)$ by induction on k. For $k = 0$, A is constant, and the claim follows by projecting its value to C. For $k \geq 1$, fix the first affine query and subtract its known constant, leaving $x \mapsto a \cdot x$. There is a Borel regular conditional-law kernel

$$y \longmapsto \nu_y \in \mathcal{P}(K) \tag{B5}$$

for ν given $a \cdot x = y$, supported on that fiber for almost every y. This standard conditional law can also be constructed directly: condition on the dyadic interval of length 2^{-m} containing y, take normalized restrictions, and then take their weak limit where it exists. Martingale convergence for a countable dense family in $C(K)$, together with compactness of $\mathcal{P}(K)$, gives that limit almost everywhere; use a fixed point mass elsewhere. This yields a Borel kernel.

For almost every y, the Borel null witness in (B1) has ν_y-measure zero. Freeze the first history. The remaining $k - 1$ stages are some well-defined adaptive algorithm on the fiber, whether or not their choice depends measurably on y. Induction applied separately to this conditional law gives

$$\int_K g \, d\nu_y \geq V_{k-1}(\nu_y). \tag{B6}$$

Enumerate $\mathcal{C}_{k-1} = (B_i)_{i \geq 1}$. For each y, choose the least $i = i(y)$ satisfying

$$\int_K \|T - B_i\|_2 \, d\nu_y \leq V_{k-1}(\nu_y) + \varepsilon. \tag{B7}$$

All quantities in (B7) are Borel in y, so this is a Borel integer-valued choice. The policy which first measures $a \cdot x$ and then follows $B_{i(y)}$ is a Borel k-query policy. Ordinary disintegration of its bounded Borel loss gives

$$\begin{aligned} V_k(\nu) &\leq \int_{\mathbb{R}} V_{k-1}(\nu_y) \, d(a_*\nu)(y) + \varepsilon \\ &\leq \int_K g \, d\nu + \varepsilon. \end{aligned} \tag{B8}$$

Let $\varepsilon \downarrow 0$, and then use (B3) once more to obtain (B2). The proof has never integrated the original nonmeasurable query choices or decoder. □

1.3 Apply the bridge to the fixed simplex lower bound

Proof of Theorem 1.1. We specify exactly which geometric facts from [1] are used. For $j \geq 1$, define

$$D_j = \sup_{\substack{f_0, \ldots, f_j \in F \\ Q : Y \to \ell_2^j, \ \|Q\| \leq 1}} \left| \det[QS(f_1 - f_0), \ldots, QS(f_j - f_0)] \right|. \tag{568.2}$$

If $D_{2n} = 0$, the affine span of $S(F)$ has dimension at most $2n - 1$. A bounded projection onto that finite-dimensional direction space then gives an exact continuous affine algorithm, and (568.1) is immediate.

Suppose $D_{2n} > 0$, and write

$$Q_n = (D_{2n}/D_n)^{1/n}. \tag{568.3}$$

Lemma 2.2 of [1] gives

$$a_{2n-1}(S, F) \le 2\sqrt{n}\, Q_n. \tag{568.4}$$

Its Lemma 2.3, Proposition 3.2, and the constant calculation in Sections 4–5 give the following stronger ingredient underlying the randomized theorem. For every $0 < \eta < 1$, one can fix a $2n$-dimensional simplex $\Delta \subset F$, a contraction $Q : Y \to \mathbb{R}^{2n}$, and an affine parametrization

$$f = \bar{f} + Vz, \quad z \in K, \qquad T = QSV, \tag{568.5}$$

where K is a compact centered simplex with its uniform probability law ν, such that every Borel finite-dimensional algorithm $\widetilde{B}$ using at most n affine measurements satisfies

$$\int_K \left\| Tz - \widetilde{B}(z) \right\|_2 d\nu(z) \ge L_\eta, \qquad L_\eta = \frac{2(1-\eta)^{1/n}}{27\sqrt{n}}\, Q_n. \tag{568.6}$$

The simplex, contraction and probability law are chosen independently of the algorithm and its random seed. Equation (568.6) is the Euclidean uniform-simplex lower bound from that proof, not an application of its global randomized theorem to a different model.

Now take an algorithm admissible in the catalogue's broader model. For each seed ω, put

$$\begin{aligned} g_\omega(z) &= \left\| S(\bar{f} + Vz) - A_\omega(\bar{f} + Vz) \right\|_Y, \\ \widetilde{A}_\omega(z) &= QA_\omega(\bar{f} + Vz) - QS\bar{f}. \end{aligned} \tag{568.7}$$

The function g_ω is Borel. To check the precise topology, the restriction of p_F to the finite-dimensional span of the simplex differences is a norm, hence is equivalent to the Euclidean norm. Thus $z \mapsto \bar{f} + Vz$ is continuous for the stipulated input topology. Joint measurability in (z, ω) follows in the same way.

The algorithm $\widetilde{A}_\omega$ uses at most n affine scalar measurements on K, and contraction gives the pointwise majorization

$$\left\| Tz - \widetilde{A}_\omega(z) \right\|_2 \le g_\omega(z). \tag{568.8}$$

The left side need not be measurable. This is why a *measurable-majorant* lemma, rather than an unqualified measurable-output replacement, was proved.

Apply Lemma 1.2 to (568.8). Since every Borel simulator is bounded below by (568.6), it follows that

$$\int_K g_\omega \, d\nu \ge L_\eta \quad \text{for every } \omega. \tag{568.9}$$

Tonelli, using only the scalar error's given joint measurability, therefore yields

$$\sup_{f \in F} \mathbb{E}_\omega \left\| Sf - A_\omega(f) \right\|_Y \ge \int_K \mathbb{E}_\omega g_\omega(z)\, d\nu(z) \ge L_\eta. \tag{568.10}$$

Take the infimum over all catalogue-admissible algorithms, let $\eta \downarrow 0$, and combine with (568.4):

$$a_{2n-1}(S, F) \le 27n\, e_n^{\mathrm{ran}}(S, F). \tag{568.11}$$

Finally, an affine algorithm of rank at most $2n - 1$ is realized by that many fixed continuous linear measurements. Monotonicity in the deterministic budget proves all inequalities in (568.1). □

References

[1] D. Krieg, E. Novak and M. Ullrich, *On the power of adaption and randomization 2: Non-symmetric convex sets*, 24 October 2026, arXiv:2610.33733v1, especially Theorem 1.1, Proposition 3.2 and its geometric proof. https://arxiv.org/abs/2610.33733.